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Roll No

EE-7102-GS
B.E. VII Semester
Examination, December 2020
Grading System (GS)
Soft Computing Techniques and Applications
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) How will you apply ANN for reliability assessments.
b) What is FLN? How FLN can be applied for identification for voltage dependent load model.
2. a) Develop an algorithm for optimization using PSO.
b) Write an expression for updating the velocity vector and position vector in PSO. Specify its each term.
3. State the techniques for proper choice of learning rate. Explain the procedure for upgradation of weights using δ - rule.
4. What is random number generation? State various methods to generate random number and explain linear congruential method with an example.
5. a) Describe the concept involved in real-coded and binary-coded genetic algorithm.
b) Explain the step-by-step procedure to optimize voltage stability problem using recombinative evolutionary strategy.

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PTO

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6. a) i) Find the constant C such that the function

$$f(x) = \begin{cases} cx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

is a density function and compute

$P(1 < x < 2)$

- ii) Find the distribution function for the random variable of $P(1 \leq x \leq 2)$. Use the result of (i) to find $P(1 < x \leq 2)$

- b) A random variable X has the density function

$$f(x) = \frac{C}{(x^2 + 1)} \text{ where, } -\infty < x < \infty$$

- i) Find the value of the constant C.
ii) Find the probability that X^2 lies between $\frac{1}{3}$ and 1.

7. Discuss the concept of RBF and FLN Network.

8. Explain distribution function and sampling distributions.
